Project Requirements

Zero-Trust Network Access: Implemented Cloudflare Tunnels and MFA to secure internal admin consoles

1. Software Dependencies

Install these packages with `python -m pip install -r requirements.txt`.

Package	Version range	Purpose
Flask	>=3.0, <4.0	Web application and routing
pyotp	>=2.9, <3.0	Time-based one-time password (TOTP) verification
gunicorn	>=22.0, <24.0	Production WSGI server (recommended for deployment)

2. Platform and Infrastructure

• Python 3.10 or newer.

• Cloudflare account with a domain/hostname managed by Cloudflare.

• cloudflared installed on the host that runs the Flask application.

• Cloudflare Zero Trust Access application configured for the admin hostname.

• Identity provider and MFA policy suitable for the organization.

• TOTP-compatible authenticator app for the application-level MFA demonstration.

• A host or VM with outbound connectivity to Cloudflare; no public inbound port is required for the origin.

3. Required Environment Variables

Variable	Description
FLASK_SECRET_KEY	Long random secret used to sign Flask sessions.
ADMIN_USERNAME	Demo administrator username.
ADMIN_PASSWORD	Strong unique demo password; never commit it.
MFA_SECRET	Base32 TOTP secret provisioned securely.
COOKIE_SECURE	Set to 1 after HTTPS/proxy behavior is reviewed.
REQUIRE_CLOUDFLARE_ACCESS_HEADER	Set to 1 only when origin access is exclusively through Cloudflare Access/Tunnel.
CF_ACCESS_EMAIL_HEADER	Header name used to display the Access identity email.

Security note: This is an academic demonstration. Cloudflare Access policy enforcement must be configured separately in Cloudflare Zero Trust. Do not use placeholder values in a live environment.